

AI Automation with n8n

Workflow Documentation

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github.com/Data-pageup/AI-Automation-n8n

This document walks through each n8n workflow in the AI Automation Lab repository: what it does, how it is structured, and how the pieces connect. A screenshot of the live n8n canvas is included under each explanation replace the placeholder image with your own exported screenshot.

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1 AI Research War Room

The **AI Research War Room** is a multi-agent research workflow that investigates a single topic from three independent angles before combining the findings into one consolidated report.

1.1 How it works

1. **Trigger & Research Brief.** The workflow starts from a manual trigger and sets up a *Research Brief* containing the topic, the guiding research question, a preferred time window for evidence, and the target audience.
2. **Three parallel search calls.** The brief fans out into three Tavily search API calls, each tuned with a different query focus:
 - *Technical Research Search* – architecture, benchmarks, papers, engineering evidence.
 - *Industry Research Search* – adoption, market trends, company case studies.
 - *Critical Research Search* – limitations, risks, criticism, counterarguments.
3. **Three independent AI agents.** Each search feeds a dedicated LLM agent (GPT-4o-mini) with a distinct persona:
 - **Technical Research Agent** – extracts concrete technical claims and separates evidence from inference.
 - **Industry Research Agent** – looks for adoption signals and business impact.
 - **Skeptic Research Agent** – stress-tests the topic, hunting for weak evidence and hype.Each agent returns a structured memo: findings, evidence, source URLs, confidence, and unknowns.
4. **Merge.** The three memos are merged in two stages (*Merge Technical + Industry*, then *Merge All Research*) into a single item stream.
5. **Lead Synthesizer.** A fourth agent, the *War Room Lead Synthesizer*, reads all three memos together, cross-checks claims across them, and produces the final report: executive summary, key findings, technical/industry/skeptical views, an evidence table, implications for the audience, recommendations, and open questions.
6. **Format Final Report.** A Code node packages the synthesized report with metadata (topic, audience, timestamp) into the final output object.

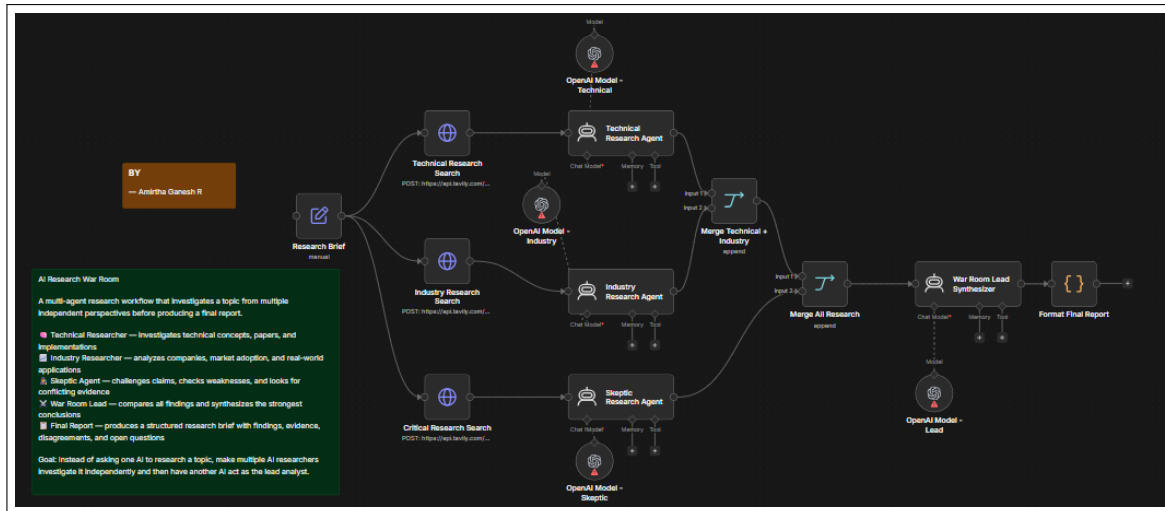


Figure 1: AI Research War Room – n8n workflow canvas

2 Autonomous Job Hunter & Resume Matcher

The **Job Hunter & Resume Matcher** workflow evaluates a job posting against a candidate profile and turns the comparison into an actionable application strategy.

2.1 How it works

1. **Input.** The workflow takes in a job posting (text or URL) and the candidate's profile/resume as inputs.
2. **Requirement extraction.** An AI step parses the job posting and extracts the explicit and implicit requirements: skills, experience level, tools, and qualifications.
3. **Profile comparison.** The extracted requirements are compared against the candidate profile to calculate a **match score** and to separate *matching skills* from *missing skills*.
4. **Gap analysis.** The workflow highlights experience gaps between what the role expects and what the candidate profile currently shows.
5. **Resume & interview guidance.** Based on the gaps identified, the workflow suggests concrete resume improvements and generates a set of interview preparation points tailored to the specific posting.
6. **Verdict.** The workflow concludes with a clear recommendation – **APPLY**, **CONSIDER**, or **SKIP** – so the output is directly actionable rather than just descriptive.

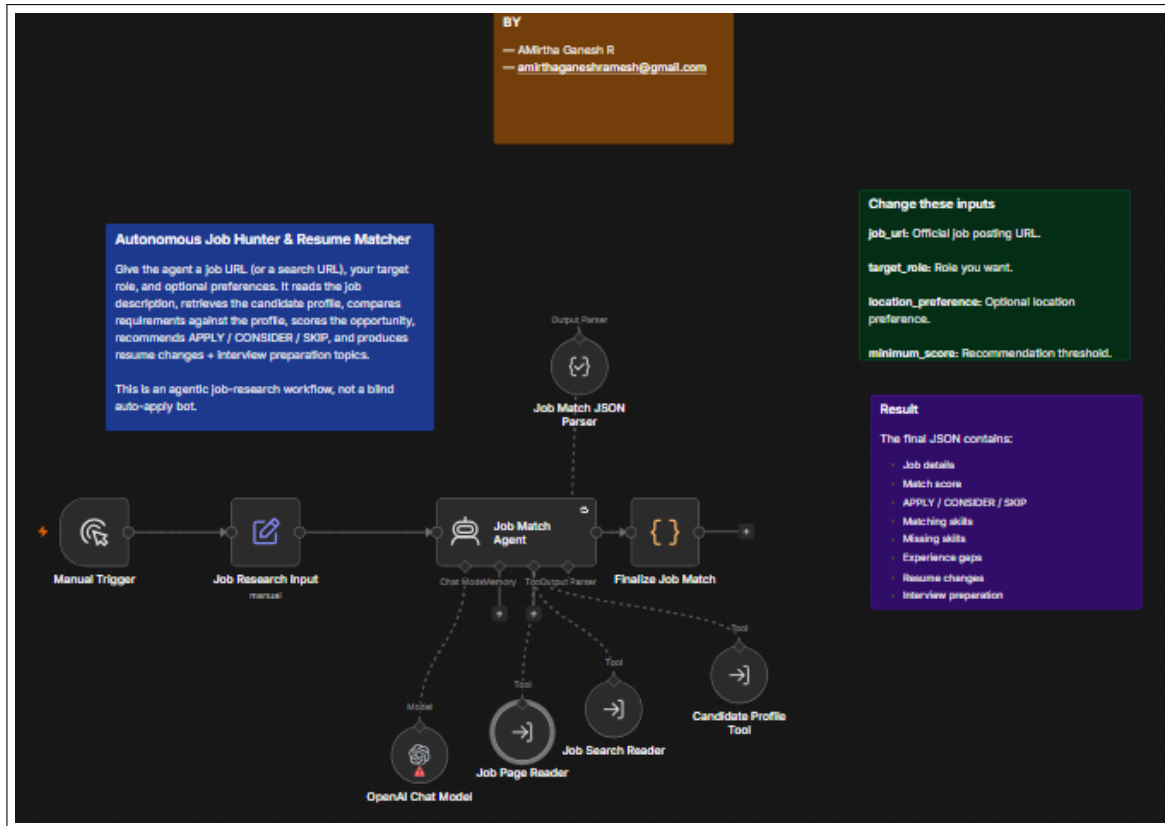


Figure 2: Autonomous Job Hunter & Resume Matcher – n8n workflow canvas

3 AI Company Intelligence Agent

The **AI Company Intelligence Agent** researches a company using its own website and other public pages, producing a structured intelligence brief.

3.1 How it works

1. **Input.** The workflow takes a company name or website URL as its starting point.
2. **Page discovery & extraction.** The workflow visits and extracts content from the company's key public pages (home, about, products, careers, etc.).
3. **Structured analysis.** An AI agent processes the extracted content and organizes it into fixed categories:
 - Company overview
 - Products and services
 - AI and technology stack
 - Target customers
 - Careers and hiring information
 - Competitors and alternatives
 - Important company pages
4. **Research takeaways.** The workflow closes with a summarized set of takeaways – the points most useful for interview prep, outreach, or general company research.

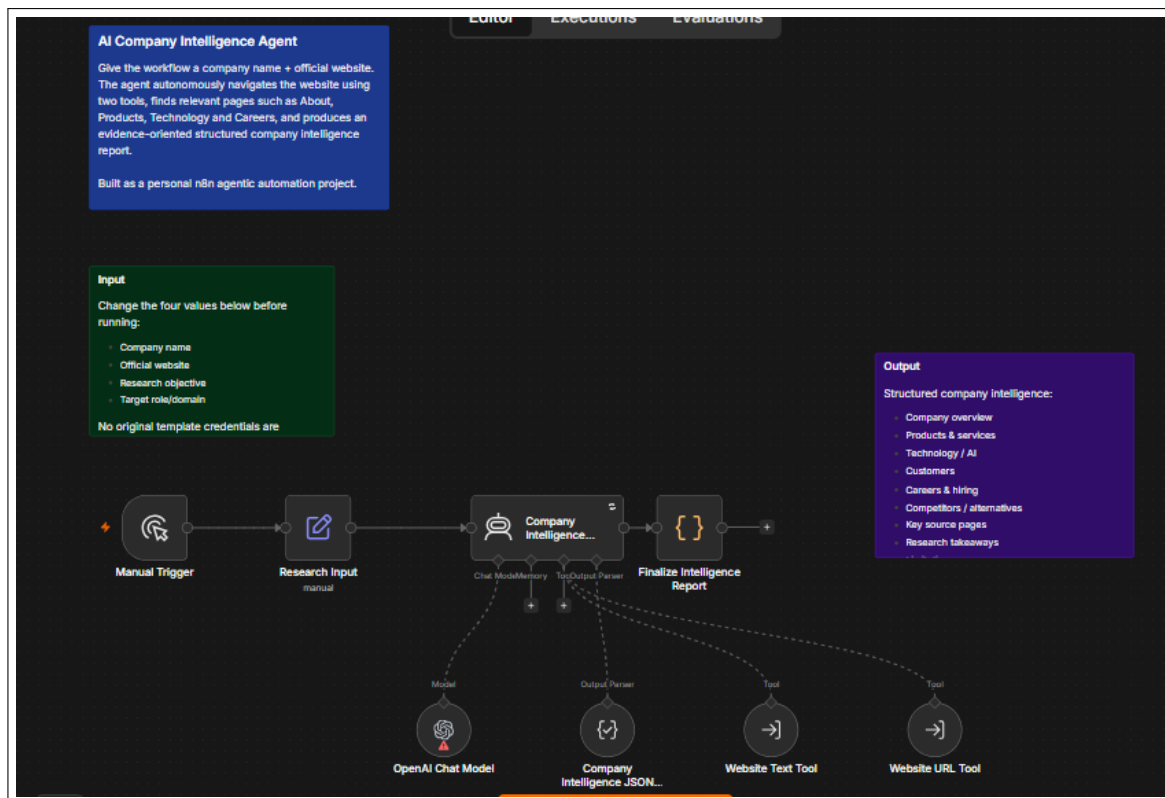


Figure 3: AI Company Intelligence Agent – n8n workflow canvas

4 Autonomous AI Crawler

The **Autonomous AI Crawler** is the foundational web-crawling workflow that the other agents build on. It discovers, cleans, and structures information from websites automatically.

4.1 How it works

1. **Crawl.** The workflow starts from one or more seed URLs and crawls linked pages within the target site.
2. **HTML extraction & Markdown conversion.** Raw HTML from each page is extracted and converted into clean Markdown, stripping navigation, ads, and boilerplate.
3. **URL processing & duplicate removal.** Discovered links are normalized and de-duplicated so the crawler does not process the same page twice.
4. **Content filtering.** Low-value or irrelevant pages are filtered out before they reach the extraction step.
5. **Structured information extraction.** The cleaned content is passed through a structuring step that outputs consistent, structured JSON – ready to be consumed by downstream workflows (such as the Company Intelligence Agent or the Research War Room).